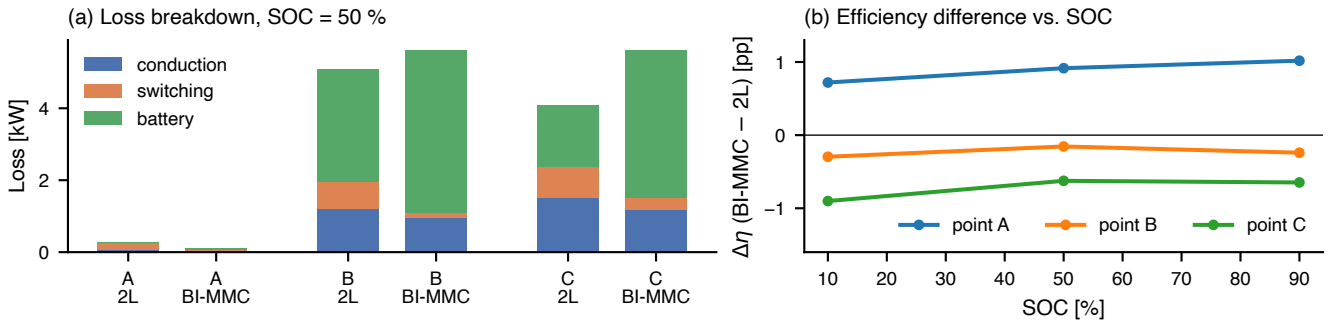
(a) Loss breakdown, SOC = 50 %
conduction
switching
battery
Loss [kW]
0
2
4
A
2L
A
BI-MMC
B
2L
B
BI-MMC
C
2L
C
BI-MMC
(b) Efficiency difference vs. SOC
Δη (BI-MMC − 2L) [pp]
1
0
−1
point A
point B
point C
10
20
30
40
50
60
70
80
90
SOC [%]